

Distortion Class Cheat Sheet

m Distortion(name, shape=None, *, display_name="", **natural)

Distortion is a factory base class: the call `Distortion('ph', 0.9)` dispatches on name to a registered subclass (PHDistortion, TVaRDistortion, ...) and returns an instance. A distortion is an increasing, concave $g : [0, 1] \rightarrow [0, 1]$ with $g(0) = 0, g(1) = 1$, used to define a coherent risk measure / pricing functional. Create one by family with a named constructor (**s** `Distortion.ph(0.9)`), or via DecL `distortion <NAME> <KIND> <SHAPE>`. The following tables show all **m** methods, **s** static / class methods, and fields or properties (used interchangeably). Comments elucidate the meaning of more obscure entries.

1. Specification & creation

name *the kind string, e.g. 'ph'*
shape *the (single) shape parameter*
kind *registry key for the subclass*
display_name *optional label override*

Named constructors, by family (each **s** static): **s** ph (proportional hazard), **s** wang (Wang / normal shift), **s** dual (dual moment), **s** power (power on $[x_0, x_1]$), **s** beta, **s** tvar (TVaR at level p), **s** bitvar ($p_0 < p_1$, weight w on p_1), **s** ccoc (constant cost of capital from discount d), **s** mean (= TVaR at $p=0$), **s** max (= TVaR at $p=1$). *Combine / generate:*
s mixture (weighted mixture), **s** minimum (pointwise min), **s** random_distortion,
s random_distortion_ex (random wtdTVaR),
s distortions_from_params (dict from a calibration table), **s** available_distortions (list registered kinds), **s** decl_spec (spec dict from a DecL number list)

2. Update

None

3. Moments

None

4. Statistical functions

m g (the distortion $g(x)$), **m** g_inv (inverse g^{-1}),
m g_dual (dual $1 - g(1 - x)$), **m** g_prime (derivative g' , central-difference default), **m** effective_g (resolve ask / bid view to the effective g), **m** make_q (vector of risk-adjusted probabilities q for pricing), **s** tvar_terms (TVaR knot function $\min(s/(1 - p), 1)$)

5. Validation

pricing_ok (the distortion yields well-defined prices),
strict_pricing (enforce strict concavity / validity),
error (numerical g vs. closed-form residual; see describe)

6. Output dataframes

info (text summary)^[1], describe^[1] (moments of g, g^{-1} vs. closed form), stats_df^[1], density_df^[1],
m tvar_info_df (affine pieces of a weighted-TVaR),
decl_params, renamer

7. Reinsurance

None

8. Visualization

m plot (g , dual, and diagonal)^[1], **m** plot_affine (upper affine envelope of a wtdTVaR)

9. Risk and pricing

m price (distorted ask / bid prices of a distribution),
m calibrate (solve shape so the distorted price hits a premium_target), premium_target, assets (capital / a), mass, has_mass (point mass $g(0^+) > 0$), gini_p (Gini coefficient)

10. Approximations

None

11. Meta

m id (short structural ID string), med_name, long_name, param_name (display labels), documented (listed by available_distortions), **m** min_index, **m** quick_gS, **m** quick_ra (internal numerical helpers)

Notes:

[1]: info, describe, stats_df, density_df, and plot mirror the common reporting interface shared with Aggregate and Portfolio — the same names do the analogous job on each class.

A distortion satisfies $g(0) = 0, g(1) = 1$, increasing and concave; the dual is $g^*(x) = 1 - g(1 - x)$. Distorted price = $\int_0^\infty g(S(x)) dx$.